

1 Summary

Model (Run-name): model1 (test)

Iteration: 3

TAC: 139792.7788895738

Took: 0.2512087821960449s

Total Time: 0.7882513999938965s

Epsilons:

- ε_{LMTD} : 1e-06
- $\varepsilon_{Balancing}$: 1e-06
- ε_{Area} : 1e-06
- ε_{Beta} : 1e-06

Gurobi Tolerances:

- IntFeasTol: 1e-05
- FeasibilityTol: 1e-06
- OptimalityTol: 1e-06

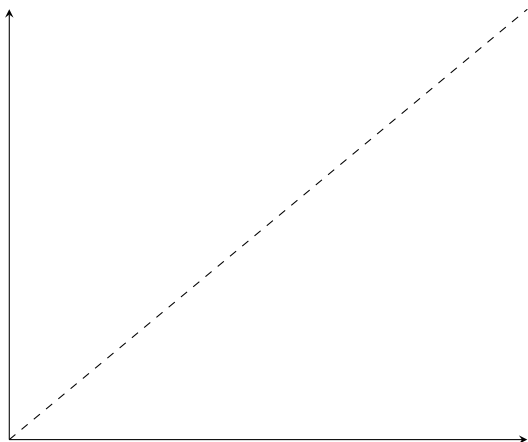
Gamsworld Solution: http://www.gamsworld.org/minlp/minlplib2/html/heatexch_gen1.html

2 Results

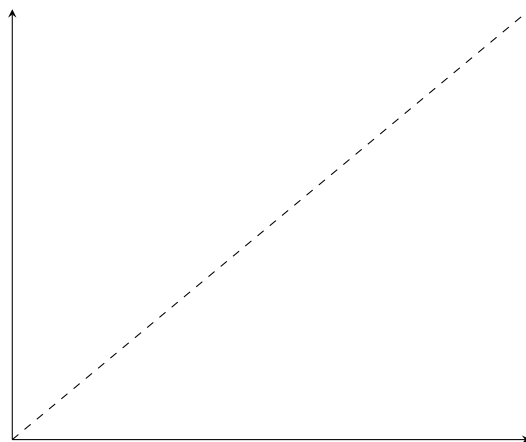
Variable	Value	Variable	Value	Variable	Value
$z_{1,1,1}$	1.0	$f_{1,2,2}^H$	9.433600312410187	$t_{1,1,1}^H$	546.9533416170123
$z_{1,2,2}$	1.0	$f_{2,1,2}^H$	17.95615118000473	$t_{1,2,2}^H$	393.58400078102545
$z_{2,1,2}$	1.0	$f_{2,2,2}^H$	2.0438488199952687	$t_{2,1,2}^H$	451.27837189025587
$z_{2,2,2}$	1.0	$f_{1,1,1}^C$	15.0	$t_{2,2,2}^H$	379.4528007924781
$z_{cu,1}$	1.0	$f_{1,2,2}^C$	11.262728503004022	$t_{1,1,1}^C$	614.9952404843539
$z_{cu,2}$	1.0	$f_{2,1,2}^C$	9.68660759081635	$t_{1,2,2}^C$	479.9545596500464
$z_{hu,1}$	1.0	$f_{2,2,2}^C$	1.7372714969959784	$t_{2,1,2}^C$	625.204168757143
$A_{1,1,1}^\beta$	60.38493803732795	$RecLMTD_{1,1,1}$	0.04927812346534701	$t_{2,2,2}^C$	370.0454403499536
$A_{1,2,2}^\beta$	11.262728503004022	$RecLMTD_{1,2,2}$	0.0151617098982103	$t_{1,1,1}^{H,in}$	3463.1588510946503
$A_{2,1,2}^\beta$	48.67452649027746	$RecLMTD_{2,1,2}$	0.018375953359037803	$t_{1,2,2}^{H,in}$	5415.681398852627
$A_{2,2,2}^\beta$	2.171589371244973	$RecLMTD_{2,2,2}$	0.012251971463148291	$t_{2,1,2}^{H,in}$	10594.129196202792
$A_{cu,1}^\beta$	7.702824927089129	$RecLMTD_{cu,1}$	0.01277087618427651	$t_{2,2,2}^{H,in}$	1205.8708037972087
$A_{cu,2}^\beta$	34.23465128608717	$RecLMTD_{cu,2}$	0.009872784857247737	$t_{1,1,1}^{H,out}$	2713.0160656252665
$A_{hu,1}^\beta$	13.30871858348944	$RecLMTD_{hu,1}$	0.02207535556803344	$t_{1,2,2}^{H,out}$	3726.2721234020237
$A_{1,1,1}$	60.38493803732795	$q_{1,1,1}$	750.1427854693839	$t_{2,1,2}^{H,out}$	8269.343374406868
$A_{1,2,2}$	11.262728503004022	$q_{1,2,2}$	1689.4092754506032	$t_{2,2,2}^{H,out}$	945.280079247812
$A_{2,1,2}$	48.67452649027746	$q_{2,1,2}$	2324.7858217959233	$t_{1,1,1}^{C,in}$	8474.785821795924
$A_{2,2,2}$	2.171589371244973	$q_{2,2,2}$	260.5907245493968	$t_{1,2,2}^{C,in}$	3941.9549760514074
$A_{cu,1}$	7.702824927089129	$q_{cu,1}$	360.4479390800125	$t_{2,1,2}^{C,in}$	3971.509112234703
$A_{cu,2}$	34.23465128608717	$q_{cu,2}$	1814.62345365468	$t_{2,2,2}^{C,in}$	608.0450239485926
$A_{hu,1}$	13.30871858348944	$q_{hu,1}$	525.0713927346924	$t_{1,1,1}^{C,out}$	9224.928607265309
$dt_{1,1,1}$	35.00475951564616	$t_{1,1}^{(H)}$	650.0	$t_{1,2,2}^{C,out}$	5631.364251502012
$dt_{1,1,2}$	10.0	$t_{1,2}^{(H)}$	574.9857214530616	$t_{2,1,2}^{C,out}$	6296.294934030626
$dt_{1,2,2}$	74.9857214530615	$t_{1,3}^{(H)}$	406.0447939080012	$t_{2,2,2}^{C,out}$	868.635748497989
$dt_{1,2,3}$	56.044793908001246	$t_{2,1}^{(H)}$	590.0		
$dt_{2,1,2}$	25.014278546938385	$t_{2,2}^{(H)}$	590.0		
$dt_{2,1,3}$	50.73117268273398	$t_{2,3}^{(H)}$	460.731172682734		
$dt_{2,2,2}$	90.0	$t_{1,1}^{(C)}$	614.9952404843539		
$dt_{2,2,3}$	70.0	$t_{1,2}^{(C)}$	564.9857214530616		
$dt_{cu,1}$	86.04479390800125	$t_{1,3}^{(C)}$	410.0		
$dt_{cu,2}$	140.73117268273398	$t_{2,1}^{(C)}$	500.0		
$dt_{hu,1}$	65.00475951564616	$t_{2,2}^{(C)}$	500.0		
$f_{1,1,1}^H$	5.32793669399177	$t_{2,3}^{(C)}$	350.0		

3 *RecLMTD* Tangent Points

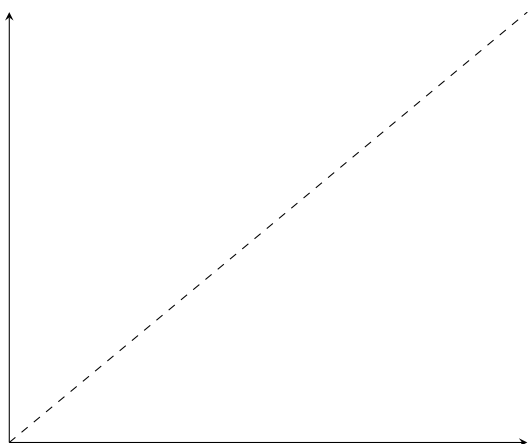
- $(1, 1, 1)$
 - Value: 0.049278
 - Error: 0.000828295059792
 - Relative error: 0.016530717703880
- $(1, 2, 2)$
 - Value: 0.015162
 - Error: 0.000209578117446
 - Relative error: 0.013634388818455
- $(2, 1, 2)$
 - Value: 0.018376
 - Error: 0.009119348368696
 - Relative error: 0.331669332418991
- $(2, 2, 2)$
 - Value: 0.012252
 - Error: 0.000313749950897
 - Relative error: 0.024968717716941
- $cu, 1$
 - Value: 0.012771
 - Error: 0.000091412928482
 - Relative error: 0.007107049739030
- $cu, 2$
 - Value: 0.009873
 - Error: 0.000000602297130
 - Relative error: 0.000061002077627
- $hu, 1$
 - Value: 0.022075
 - Error: 0.000014872159939
 - Relative error: 0.000673246112691



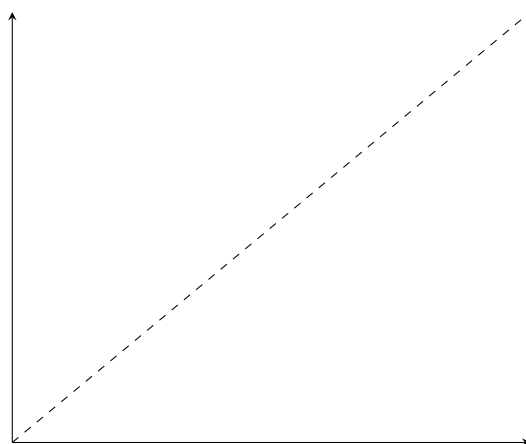
(a) *RecLMTD* - (1, 1, 1)



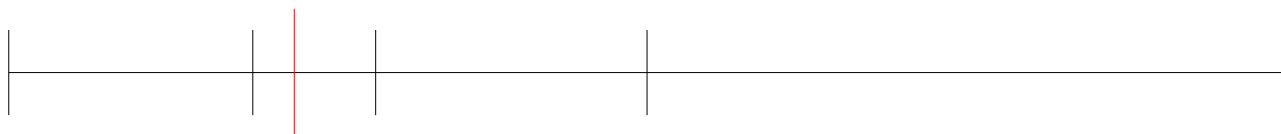
(b) *RecLMTD* - (1, 2, 2)



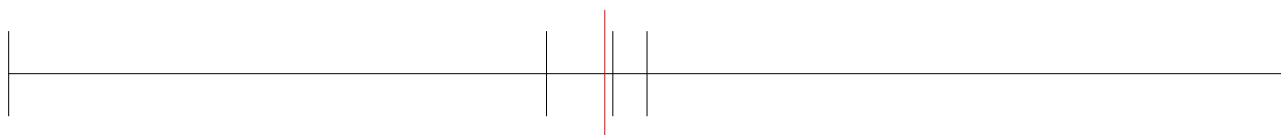
(c) *RecLMTD* - (2, 1, 2)



(d) *RecLMTD* - (2, 2, 2)



(a) *RecLMTD* - *cu*, 1



(b) *RecLMTD* - *cu*, 2



(a) *RecLMTD* - *hu*, 1

4 Balancing Breakpoints

4.1 $b^{H,\text{in}}$

- (1, 1, 1)
 - Value: 3463.1588510946503
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 2, 2)
 - Value: 5415.681398852627
 - Absolute error: 8.50408267837156
 - Relative error: -0.001567808237260214
- (2, 1, 2)
 - Value: 10594.129196202792
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 2, 2)
 - Value: 1205.8708037972087
 - Absolute error: 2.2737367544323206e-13
 - Relative error: 1.8855558549659482e-16

4.2 $b^{H,\text{out}}$

- (1, 1, 1)
 - Value: 2713.0160656252665
 - Absolute error: 201.1167130774288
 - Relative error: -0.06901425856338685
- (1, 2, 2)
 - Value: 3726.2721234020237
 - Absolute error: 13.357970674490844
 - Relative error: 0.0035977052323383197
- (2, 1, 2)
 - Value: 8269.343374406868
 - Absolute error: 166.12070447903625
 - Relative error: 0.02050057258028129
- (2, 2, 2)
 - Value: 945.280079247812
 - Absolute error: 169.73592010420577
 - Relative error: 0.21886041962025282

4.3 $b^{C,in}$

- (1, 1, 1)
 - Value: 8474.785821795924
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 2, 2)
 - Value: 3941.9549760514074
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 2)
 - Value: 3971.509112234703
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 2, 2)
 - Value: 608.0450239485926
 - Absolute error: 2.2737367544323206e-13
 - Relative error: 3.7394216955627207e-16

4.4 $b^{C,out}$

- (1, 1, 1)
 - Value: 9224.928607265309
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 2, 2)
 - Value: 5631.364251502012
 - Absolute error: 225.76635238469044
 - Relative error: 0.04176528787343871
- (2, 1, 2)
 - Value: 6296.294934030626
 - Absolute error: 240.18748713765945
 - Relative error: 0.03966037413369962
- (2, 2, 2)
 - Value: 868.635748497989
 - Absolute error: 225.76635238468907
 - Relative error: 0.35118540989762687



(a) $t^{(H)} - (1, 1)$



(a) $t^{(H)} - (1, 2)$



(a) $t^{(H)} - (2, 2)$



(a) $t^{(H)} - (2, 2)$



(a) $t^{(H)} - (1, 1, 1)$



(a) $t^{(H)} - (1, 2, 2)$



(a) $t^{(H)} - (2, 1, 2)$



(a) $t^{(H)} - (2, 2, 2)$



(a) $t^{(C)} - (1, 2)$



(a) $t^{(C)} - (2, 3)$



(a) $t^{(C)} - (1, 3)$



(a) $t^{(C)} - (2, 3)$



(a) $t^{(C)} - (1, 1, 1)$



(a) $t^{(C)} - (1, 2, 2)$



(a) $t^{(C)} - (2, 1, 2)$



(a) $t^{(C)} - (2, 2, 2)$

5 Area Breakpoints

5.1 A

- $(1, 1, 1)$
 - Value: 60.384938
 - Error: 13.546319560671272
 - Relative error: 0.183228582886136
- $(1, 2, 2)$
 - Value: 11.262729
 - Error: 39.965938164451387
 - Relative error: 0.780147928188047
- $(2, 1, 2)$
 - Value: 48.674526
 - Error: 36.765785171871059
 - Relative error: 0.430309586384139
- $(2, 2, 2)$
 - Value: 2.171589
 - Error: 4.213910870235720
 - Relative error: 0.659918676826889

5.2 A_{CU}

- $cu, 1$
 - Value: 7.702825
 - Error: 1.503647074647834
 - Relative error: 0.163325003797779
- $cu, 2$
 - Value: 34.234651
 - Error: 1.596122623609858
 - Relative error: 0.044546138680468

5.3 A_{HU}

- $hu, 1$
 - Value: 13.308719
 - Error: 0.600646641299676
 - Relative error: 0.043182893798001



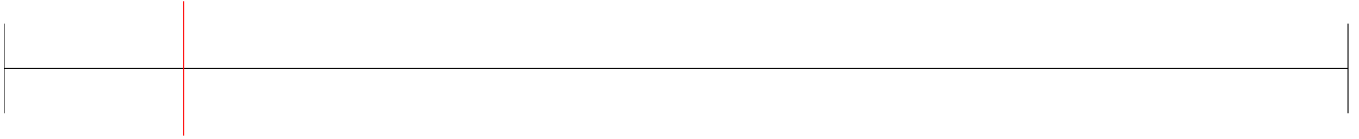
(a) $A^\beta - (1, 1, 1)$



(a) $A^\beta - (1, 2, 2)$



(a) $A^\beta - (2, 1, 2)$



(a) $A^\beta - (2, 2, 2)$



(a) $A^\beta - cu, 1$



(a) $A^\beta - cu, 2$

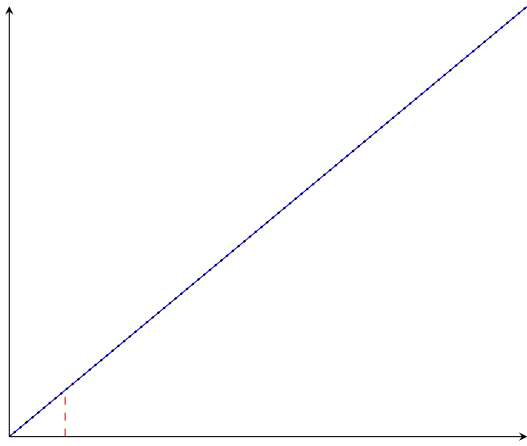


(a) $A^\beta - hu, 1$

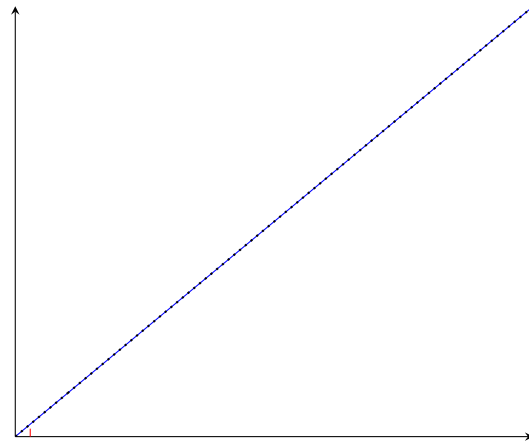
6 A^β Breakpoints

- (1,1,1)
 - q : 60.384938
 - $\hat{q}^\beta$: 60.384938
 - q^β : 60.384938
 - Breakpoints: [0, 5.0, 18.13353447520507, 560.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- (1,2,2)
 - q : 11.262729
 - $\hat{q}^\beta$: 11.262729
 - q^β : 11.262729
 - Breakpoints: [0, 39.36524453694061, 390.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- (2,1,2)
 - q : 48.674526
 - $\hat{q}^\beta$: 48.674526
 - q^β : 48.674526
 - Breakpoints: [0, 9.898970403313276, 28.33333333333332, 720.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- (2,2,2)
 - q : 2.171589
 - $\hat{q}^\beta$: 2.171589
 - q^β : 2.171589
 - Breakpoints: [0, 38.943261778500734, 390.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $cu, 1$
 - q : 7.702825
 - $\hat{q}^\beta$: 7.702825
 - q^β : 7.702825
 - Breakpoints: [0, 2.9819181007907165, 9.28003023700696, 181.61828057849593]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $cu, 2$

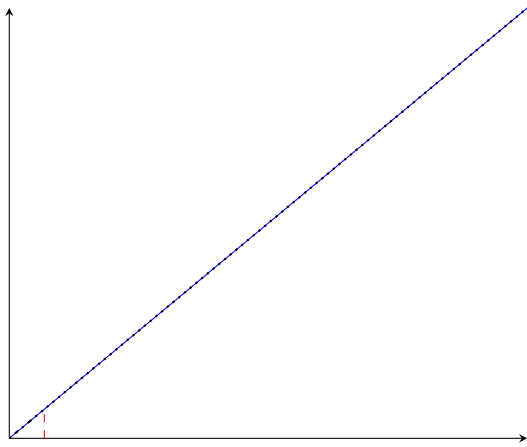
- q : 34.234651
- $\hat{q}^\beta$: 34.234651
- q^β : 34.234651
- Breakpoints: [0, 21.04660840930274, 24.97364426355679, 285.4001551947793]
- Error: 0.0000000000000000
- Relative error: 0.0000000000000000
- $hu, 1$
 - q : 13.308719
 - $\hat{q}^\beta$: 13.308719
 - q^β : 13.308719
 - Breakpoints: [0, 4.943755299006499, 12.595096067947068, 237.3002543523117]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000



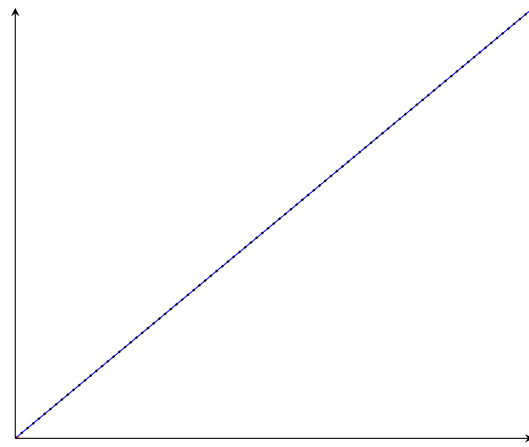
(a) $q^\beta - (1, 1, 1)$



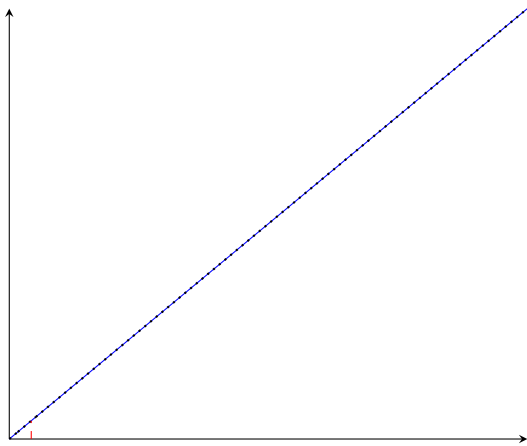
(b) $q^\beta - (1, 2, 2)$



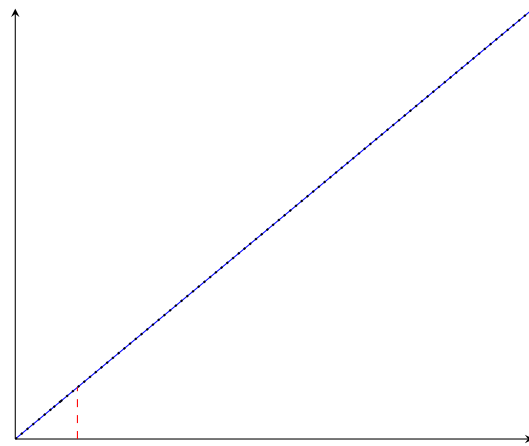
(c) $q^\beta - (2, 1, 2)$



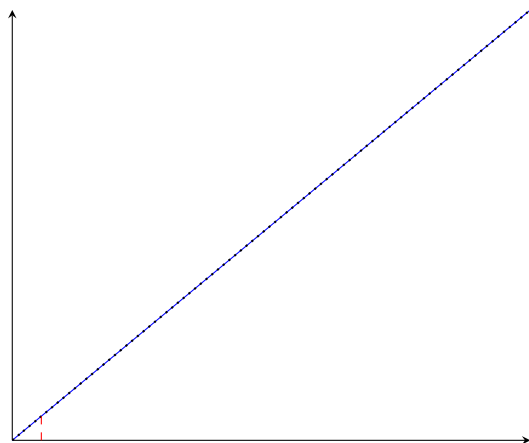
(d) $q^\beta - (2, 2, 2)$



(a) $q^\beta - cu, 1$



(b) $q^\beta - cu, 2$



7 Other variables

Variable	Value	Variable	Value
$z_{1,1,2}$	0.0	$q_{1,1,2}$	0.0
$z_{1,2,1}$	-0.0	$q_{1,2,1}$	0.0
$z_{2,1,1}$	-0.0	$q_{2,1,1}$	0.0
$z_{2,2,1}$	0.0	$q_{2,2,1}$	0.0
$z_{hu,2}$	-0.0	$q_{hu,2}$	0.0
$A_{1,1,2}^\beta$	0.0	$\sigma_{1,1,2}^H$	436.91512735787
$A_{1,2,1}^\beta$	0.0	$\sigma_{1,2,1}^H$	650.0
$A_{2,1,1}^\beta$	0.0	$\sigma_{2,1,1}^H$	590.0
$A_{2,2,1}^\beta$	0.0	$\sigma_{2,2,1}^H$	590.0
$A_{hu,2}^\beta$	0.0	$\sigma_{1,1,2}^C$	410.0
$A_{1,1,2}$	0.0	$\sigma_{1,2,1}^C$	500.0
$A_{1,2,1}$	0.0	$\sigma_{2,1,1}^C$	580.0
$A_{2,1,1}$	0.0	$\sigma_{2,2,1}^C$	500.0
$A_{2,2,1}$	0.0	$\sigma_{1,1,2}^{H,in}$	334.1758156779888
$A_{hu,2}$	0.0	$\sigma_{1,2,1}^{H,in}$	3036.8411489053497
$dt_{1,1,3}$	10.0	$\sigma_{2,1,1}^{H,in}$	11800.0
$dt_{1,2,1}$	280.0	$\sigma_{2,2,1}^{H,in}$	0.0
$dt_{2,1,1}$	10.0	$\sigma_{1,1,2}^{H,out}$	334.1758156779888
$dt_{2,2,1}$	220.0	$\sigma_{1,2,1}^{H,out}$	3036.8411489053497
$dt_{hu,2}$	180.0	$\sigma_{2,1,1}^{H,out}$	11800.0
$f_{1,1,2}^H$	0.5663996875898132	$\sigma_{2,2,1}^{H,out}$	0.0
$f_{1,2,1}^H$	4.67206330600823	$\sigma_{1,1,2}^{C,in}$	2178.490887765297
$f_{2,1,1}^H$	20.0	$\sigma_{1,2,1}^{C,in}$	68.75855297536054
$f_{2,2,1}^H$	0.0	$\sigma_{2,1,1}^{C,in}$	0.0
$f_{1,1,2}^C$	5.313392409183651	$\sigma_{2,2,1}^{C,in}$	6431.2414470246395
$f_{1,2,1}^C$	0.13751710595072097	$\sigma_{1,1,2}^{C,out}$	2178.490887765298
$f_{2,1,1}^C$	0.0	$\sigma_{1,2,1}^{C,out}$	68.75855297536054
$f_{2,2,1}^C$	12.862482894049279	$\sigma_{2,1,1}^{C,out}$	0.0
$RecLMTD_{1,1,2}$	0.1	$\sigma_{2,2,1}^{C,out}$	6431.2414470246395
$RecLMTD_{1,2,1}$	0.01607047381606835		
$RecLMTD_{2,1,1}$	0.1		
$RecLMTD_{2,2,1}$	0.006080000000000005		
$RecLMTD_{hu,2}$	0.005549329826813255		